

Name of Student: Dhruv Sharma	PRN:25070122075
Semester: III	Year : AY 2026-27
Subject: Programming with Java	
ASSIGNMENT No : 2	
TITLE : Constructors in Java	

Problem Statement

Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Learning Objectives

To understand and implement default, parameterized, and copy constructors for object initialization.

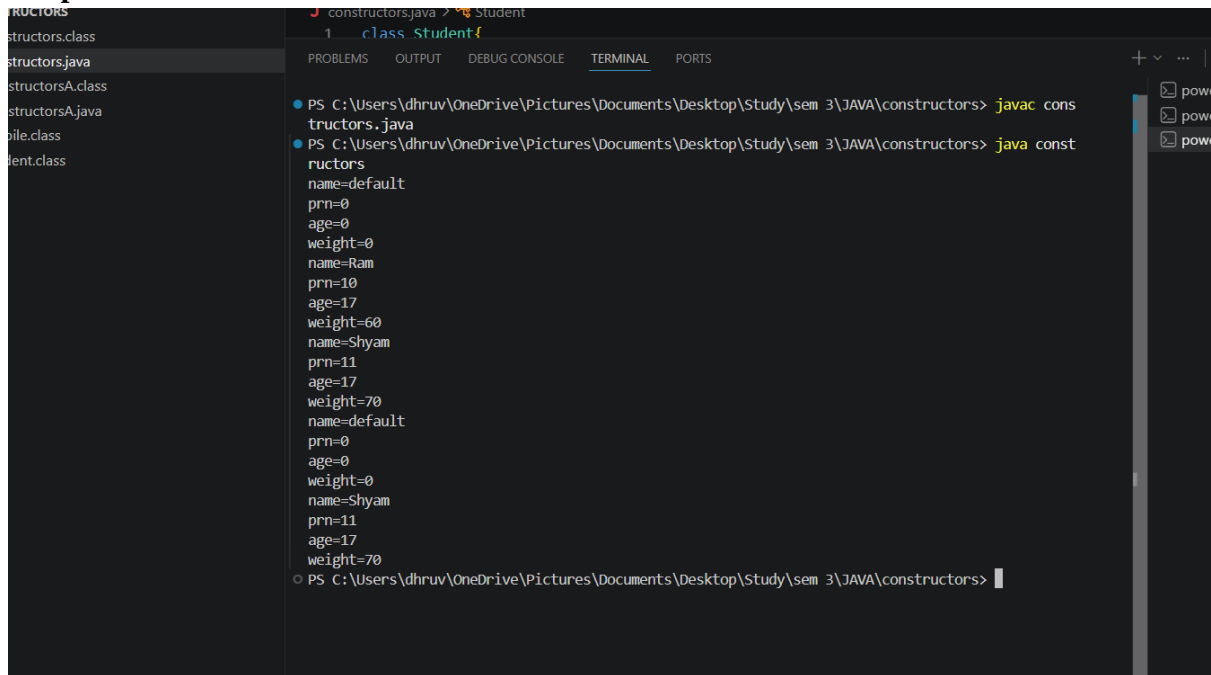
Programme

QUESTIONS-

1. 1. Create a Student class using default ,parameterized and copy constructors to initialize student name and roll number.

```
J constructors.java X
J constructors.java > Student > Student()
1 class Student{
2     String name;
3     int prn;
4     int weight;
5     int age;
6
7     Student(){ // a default constructor is constructed here
8         name="default";
9         age=0;
10        prn=0;
11        weight=0;
12    }
13
14    Student(String n , int p ,int a , int wg ){ // parameterized constructor is called here
15        name=n;
16        prn=p;
17        age=a;
18        weight=wg;
19    }
20
21    Student(Student copySt){ //here copy constructor is called (here object reference is used , reference acts like pointer and also holds memory address of object)
22        this.name=copySt.name;
23        this.prn=copySt.prn;
24        this.age=copySt.age;
25        this.weight=copySt.weight;
26    }
27
28    void display(){
29        System.out.println("name="+name);
30        System.out.println("prn="+prn);
31        System.out.println("age="+age);
32        System.out.println("weight="+weight);
33    }
34 }
35
36 public class constructors{
37     Run main | Debug main
38     public static void main(String[] args) {
39         Student s1 = new Student(); //default constructor is called
40         Student s2 = new Student("Ram",10,17,60); //parameterised is called
41         Student s3 = new Student("Shyam",11,17,70); //parameterised constructor is called
42         Student s4 = new Student(s1); //copy constructor is called
43         Student s5 = new Student(s3); //copy constructor is called
44
45         s1.display();
46         s2.display();
47         s3.display();
48         s4.display();
49         s5.display();
50     }
51 }
```

Output-



The screenshot shows an IDE with a file explorer on the left, a code editor in the center, and a terminal at the bottom. The file explorer lists files: constructors.class, constructors.java, constructorsA.class, constructorsA.java, mobile.class, and student.class. The code editor shows the following Java code:

```
1 class Student{
```

The terminal displays the output of two commands:

```
PS C:\Users\dhruv\OneDrive\Pictures\Documents\Desktop\Study\sem 3\JAVA\constructors> javac constructors.java
PS C:\Users\dhruv\OneDrive\Pictures\Documents\Desktop\Study\sem 3\JAVA\constructors> java constructors
name=default
prn=0
age=0
weight=0
name=Ram
prn=10
age=17
weight=60
name=Shyam
prn=11
age=17
weight=70
name=default
prn=0
age=0
weight=0
name=Shyam
prn=11
age=17
weight=70
PS C:\Users\dhruv\OneDrive\Pictures\Documents\Desktop\Study\sem 3\JAVA\constructors>
```

2. Develop a Mobile Phone Inventory System using different constructors to initialize mobile details and create duplicate object records.

Code-

```
Selection View Go Run Terminal Help constructors
J ConstructorsA.java X
J ConstructorsA.java > ...
1 // question - 2. Develop a Mobile Phone Inventory System using different constructors to initialize mobile details and create duplicate object records.
2
3
4 class Mobile {
5     String brand;
6     String model;
7     int price;
8     int quantity;
9
10    Mobile(String b, String m, int p, int q) { // Parameterized Constructor
11        brand = b;
12        model = m;
13        price = p;
14        quantity = q;
15    }
16
17    Mobile(Mobile copyMb) { // Copy Constructor
18        this.brand = copyMb.brand;
19        this.model = copyMb.model;
20        this.price = copyMb.price;
21        this.quantity = copyMb.quantity;
22    }
23
24    void display() {
25        System.out.println("Brand = " + brand);
26        System.out.println("Model = " + model);
27        System.out.println("Price = " + price);
28        System.out.println("Quantity = " + quantity);
29        System.out.println();
30    }
31 }
32
33
34 public class ConstructorsA {
35     Run main | Debug main
36     public static void main(String[] args) {
37
38         // Parameterized Constructor
39         Mobile m1 = new Mobile("Samsung", "Galaxy S24", 75000, 10);
40         Mobile m2 = new Mobile("Apple", "iPhone 15", 80000, 5);
41
42         // Copy Constructor
43         Mobile m3 = new Mobile(m1);
44         Mobile m4 = new Mobile(m2);
45
46         System.out.println("Original Mobile 1:");
47         m1.display();
48
49         System.out.println("Original Mobile 2:");
50         m2.display();
51
52         System.out.println("Duplicate Mobile 1:");
53         m3.display();
54
55         System.out.println("Duplicate Mobile 2:");
56         m4.display();
57     }
58 }
```

Output-

```
Constructors.java
ConstructorsA.class
ConstructorsA.java C:\Users\dhruv\OneDrive\Pictures\Documents\Desktop\Study\sem
3\JAVA\constructors\ConstructorsA.class Desktop\Study\sem 3\JAVA\constructors> javac Constructo
ConstructorsA.class PS C:\Users\dhruv\OneDrive\Pictures\Documents\Desktop\Study\sem 3\JAVA\constructors> java Constructor
SA
Original Mobile 1:
Brand = Samsung
Model = Galaxy S24
Price = 75000
Quantity = 10

Original Mobile 2:
Brand = Apple
Model = iPhone 15
Price = 80000
Quantity = 5

Duplicate Mobile 1:
Brand = Samsung
Model = Galaxy S24
Price = 75000
Quantity = 10

Duplicate Mobile 2:
Brand = Apple
Model = iPhone 15
Price = 80000
Quantity = 5

PS C:\Users\dhruv\OneDrive\Pictures\Documents\Desktop\Study\sem 3\JAVA\constructors>
```

Conclusion: From this assignment we get to know about constructors and its type. Also learn how they are used in Java Program and how they are different from each other.